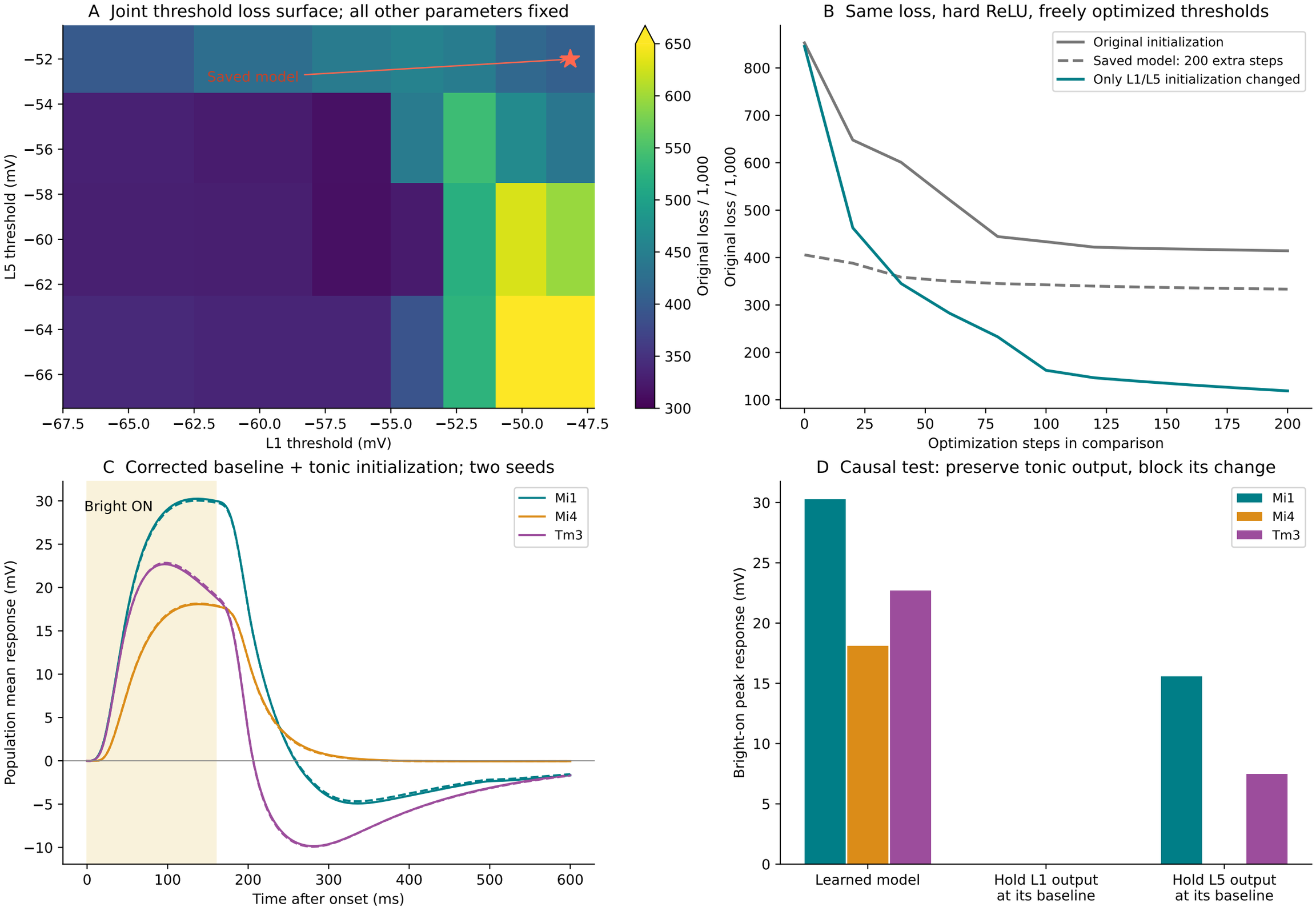
A Joint threshold loss surface; all other parameters fixed
Saved model
L5 threshold (mV)
L1 threshold (mV)
Original loss / 1,000
B Same loss, hard ReLU, freely optimized thresholds
Original initialization
Saved model: 200 extra steps
Only L1/L5 initialization changed
Original loss / 1,000
Optimization steps in comparison
C Corrected baseline + tonic initialization; two seeds
Bright ON
Mi1
Mi4
Tm3
Population mean response (mV)
Time after onset (ms)
D Causal test: preserve tonic output, block its change
Bright-on peak response (mV)
Learned model
Hold L1 output at its baseline
Hold L5 output at its baseline